

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATIONS  
Higher 2

CANDIDATE  
NAME

CLASS

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**BIOLOGY**

Paper 3 Applications and Planning

**9648/03**

**23 September 2014**

**2 hours**

Additional Materials: Answer Paper

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**READ THESE INSTRUCTIONS FIRST**

Write your name and CT on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

| For Examiner's Use |           |
|--------------------|-----------|
| 1                  |           |
| 2                  |           |
| 3                  |           |
| 5                  |           |
| Total              | <b>60</b> |
| For Examiner's Use |           |
| 4                  | <b>12</b> |

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This document consists of **16** printed pages and **0** blank pages.

**[Turn over**

### Section A

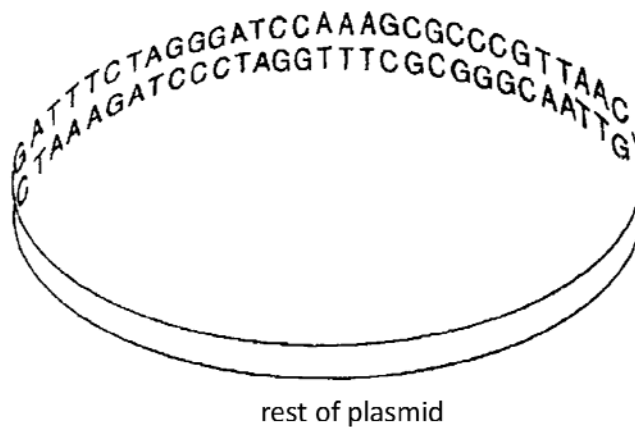
Answer **all** the questions in this section.

- 1 Restriction enzymes cut DNA molecules only at specific target sites with particular base sequences. The target sites for the enzymes *EcoRI*, *BamHI* and *HpaI* are shown in **Table 1.1**. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

**Table 1.1**

| restriction enzyme | specific target base sequence of DNA                             |
|--------------------|------------------------------------------------------------------|
| <i>EcoRI</i>       | <pre> G   A A T T C               C T T A A   G           </pre> |
| <i>BamHI</i>       | <pre> G   G A T C C             C C T A G   G           </pre>   |
| <i>HpaI</i>        | <pre> G T T   A A C             C A A   T T G           </pre>   |

A bacterial plasmid includes the base sequence shown in **Fig. 1.1**. There are no target sequences for these three restriction enzymes in the rest of the plasmid.



**Fig. 1.1**

- (a) State how many restriction fragments of DNA will be present after the plasmid has been treated with.
- (i) *HpaI* and *EcoRI*;

..... [1]

- (ii) *HpaI* and *BamHI*;

..... [1]

- (b) A genetic engineer cuts the plasmid shown in **Fig. 1.1** with *HpaI* in order to insert a human gene.
- (i) Describe the steps that must be subsequently followed to produce a recombinant plasmid.

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[3]

- (ii) Describe any difference in the procedure that would be needed to produce a recombinant plasmid if the plasmid were cut with *Bam*HI instead of *HpaI*.

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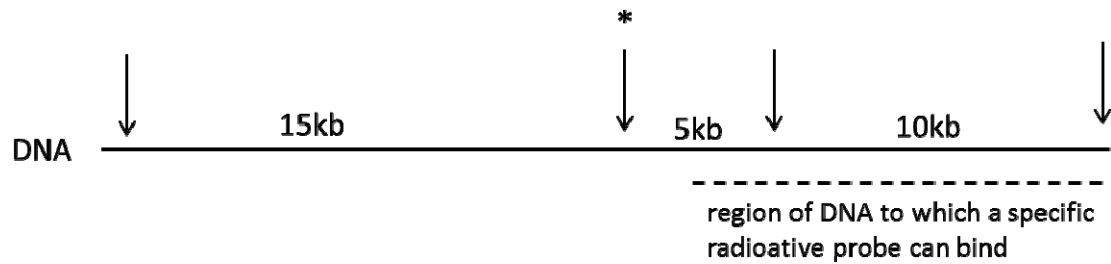
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[2]

A length of DNA from one of a pair of homologous chromosomes is shown in **Fig. 1.2**. The target sites of EcoRI are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

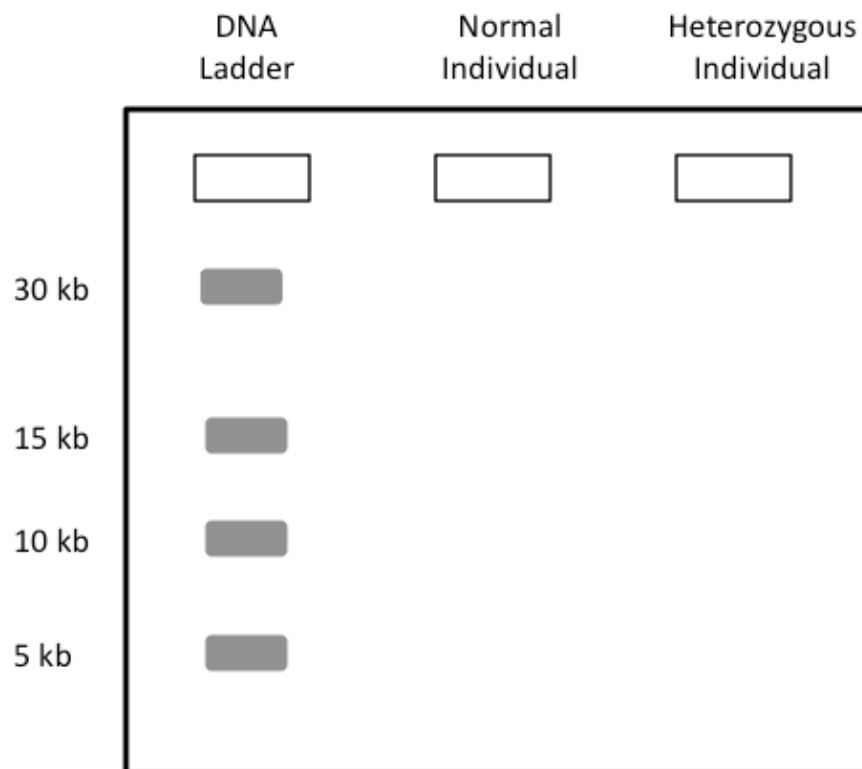


**Fig. 1.2**

A mutation alters one base of the coding of the site marked with an asterisk (\*). DNA from an individual who is heterozygous for this mutation is cut with EcoRI and the DNA fragments separated by electrophoresis.

(c) On **Fig. 1.3**,

- (i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individuals tested and label the size of the fragments produced.



**Fig. 1.3**

[3]

- (ii) Circle the fragments which will combine with the radioactive probe and appear as black bands on an autoradiography;

[2]

- (d) DNA fragments are separated by electrophoresis in genetic fingerprinting. Briefly describe how DNA fragments may be separated by electrophoresis.

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[3]

[Total: 15]

- 2 The first gene therapy trial for an inherited disorder was initiated on 14 September 1990. The trial was for a patient named Ashanthi DeSilva, who suffered from a very rare recessively inherited disorder, adenosine deaminase (ADA) deficiency.

An inherited deficiency of ADA has particularly severe consequences in the case of T lymphocytes. As a result, ADA-deficient patients suffer from severe combined immunodeficiency (SCID).

- (a) Explain how a mutation of ADA gene affects the immune system of ADA-deficient SCID patients.

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- (b) Suggest why a recessively inherited disorder such as ADA-SCID can be treated by gene therapy. [3]

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- (c) The novel ADA gene therapy approach involved essentially the following steps: [2]

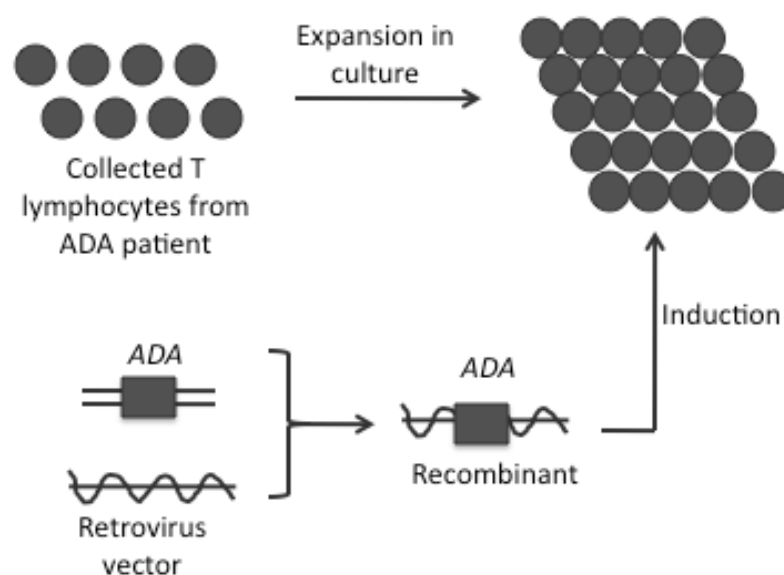


Fig. 2.1

- (i) Suggest why the *ex vivo* method was used in this case.

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- (ii) Suggest an improvement to the procedure in **Fig. 2.1** so as to increase the success rate of the treatment. [1]

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- (d) Gene therapy using the same form of treatment as the above case study has succeeded for 15 French baby boys suffering from X-linked SCID. However, three of the little boys later developed cancer and died. Suggest how this is possible. [3]

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- (e) State and explain another factor that hinders the effectiveness of current gene therapy approaches. [2]

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[2]  
[Total: 13]

- 3** Deficiency of vitamin A in the diet is a major problem in the rice-based societies of South-East Asia. Rice plants produce the precursor of vitamin A,  $\beta$ -carotene, in their green tissues, but not in the edible endosperms of their seeds.

Rice seeds lack the enzymes for two steps of the pathway for  $\beta$ -carotene production. The genes for these two enzymes have been inserted into rice embryos by genetic engineering, giving rise to Golden Rice™. This rice produces seeds containing  $\beta$ -carotene.

The added genes were:

- the *psy* gene from daffodil plants,
- the *crt 1* gene from the bacterium *Erwinia uredovora*.

- (a)** Explain why it is possible for rice plants to express genes from a bacterium.

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.....

[1]

The following steps were taken to produce Golden Rice™:

- **Step 1:**  
A length of DNA that included a rice endosperm-specific promoter, *psy*, *crt 1* was produced.
- **Step 2:**  
Copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*.
- **Step 3:**  
*A. tumefaciens* containing the plasmids were mixed with rice embryos in tissue culture
- **Step 4:**  
The embryos were grown into plantlets and then plants.

- (b)** Describe the role of the rice endosperm-specific promoter that was added to *psy* and *crt 1* in **step 1**.

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[2]



The concentration of  $\beta$ -carotene was increased by further genetic engineering. More recent strains, such as Golden Rice 2™, contain about  $31 \mu\text{g g}^{-1}$  of  $\beta$ -carotene in the endosperm of their seeds.

- (c) Suggest what further genetic engineering was carried out to increase the concentration of  $\beta$ -carotene in rice seeds.

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 ..... [1]

- (d) Field trials of genetically engineered crops, such as Golden Rice™, hope to identify any risks to the environment or to human health of growing and eating the crops. Suggest one possible risks to the environment of growing genetically engineered crops;

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 ..... [1]

In another variety of rice plant, an enzyme, acetolactate synthase (ALS) catalyse the first step in the biosynthesis of valine and isoleucine. A common herbicide used by farmers, chlorsulfuron, works by binding to ALS. Thus, a single amino acid change in the ALS protein confers the plant resistance to the herbicide chlorsulfuron.

In some rice plants, the allele that results in abnormal ALS protein was identified. The presence of this allele makes the rice plant naturally resistant to the herbicide.

This allele was inserted into a vector and introduced into *Agrobacterium tumefaciens*. Rice protoplasts were transformed with *A. tumefaciens* carrying this vector and the transformed protoplasts could be stimulated to divide by mitosis to form calli and regenerated into whole plants by tissue culture techniques.

- (e) With reference to tissue culture techniques, describe how plant growth regulators can result in plantlet formation from calli.

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 ..... [2]

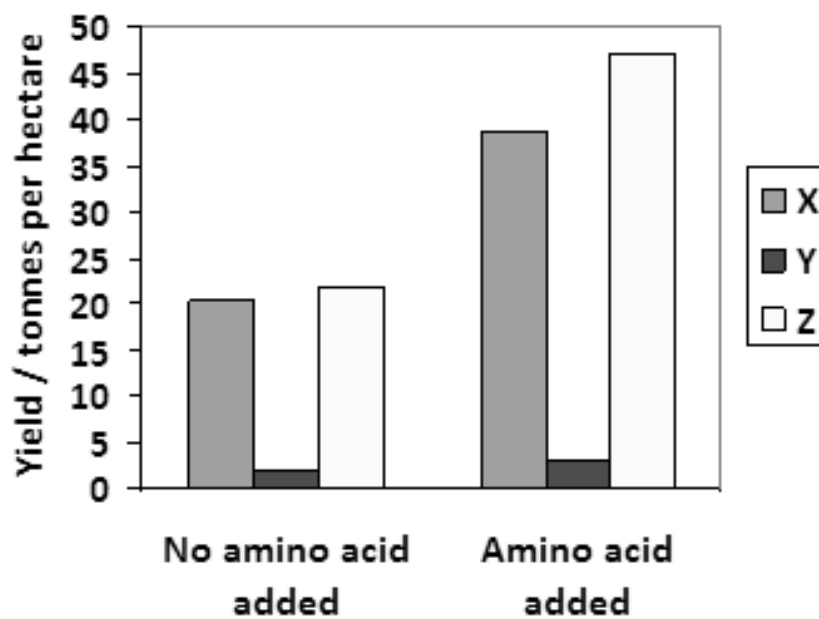
The effectiveness of abnormal ALS protein was tested using two groups of rice plants. One group was given fertilizer that contains valine and isoleucine while the other group was given fertilizer which does not contain valine and isoleucine.

Each group of rice plants consists of plant type X, Y and Z and both groups are subjected to identical growing conditions and to herbicide spraying at regular intervals.

**X:** Susceptible rice plants genetically engineered, as described above to express the allele coding for abnormal ALS protein

**Y:** Susceptible rice plants

**Z:** Resistant rice plants



**Fig. 3.1**

- (f) With reference to Fig. 3.1,  
 (i) describe and explain the effects of herbicides on yield of plant type X and Z when amino acids are added.

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[2]

(ii) explain the effects of amino acids to plant types X and Z.

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[3]

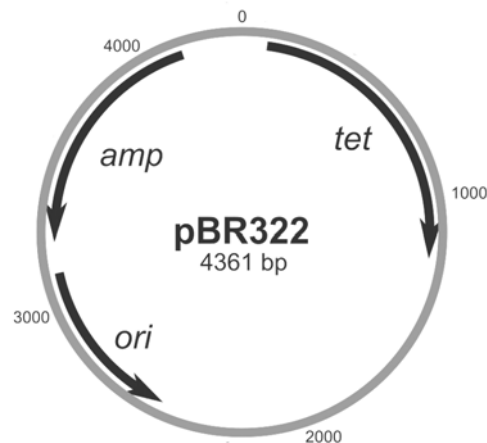
[Total: 12]

#### 4 Planning Question

Electroporation is a method that is used to introduce foreign DNA into *E. coli* cells. This involves subjecting a mixture of bacteria and plasmid to a brief electric pulse.

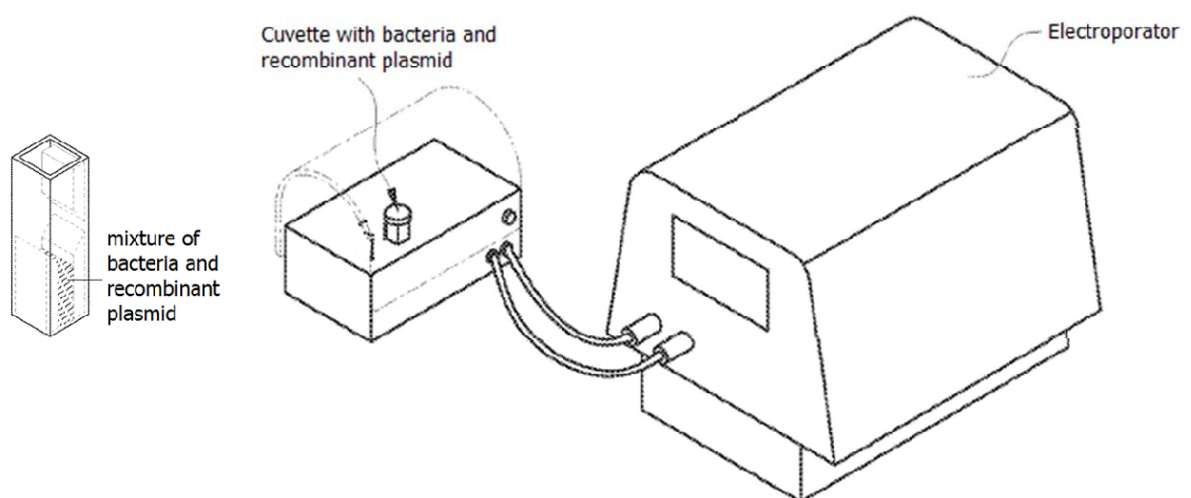
You are required to plan, but not carry out, an investigation into the effect of voltage on the efficiency of transformation by electroporation.

In this investigation, pBR322 (**Fig. 4.1**) was used to transform the *E. coli* cells. pBR322 contains both ampicillin and tetracycline resistance genes.



**Fig. 4.1**

An electroporator is used to carry out electroporation. The voltage and length of the electrical pulse can be varied with this machine. The range of voltage used is typically 1400 – 2000 V and the length of electrical impulse between 4-6 ms. A voltage which is too high will cause irreversible changes in membrane permeability. The mixture of bacteria and plasmid, which must be kept on ice prior to electroporation, is put into a cuvette which is then placed into a chamber in the electroporator (**Fig. 4.2**).



**Fig. 4.2**

After electroporation, the mixture of bacteria must be quickly resuspended in SOC broth (a type of growth medium) and incubated at 37°C for 1 hour for the surviving cells to recover. This is then followed by plating the bacteria on agar and incubation at 37°C for 16-24 hours. The plates can then be examined to calculate transformation efficiency.

Transformation efficiency can be calculated using the following equation:

$$\text{Transformation efficiency / cfu } \mu\text{g}^{-1} = \frac{\text{Total number of colonies on agar plate}}{\text{Amount of DNA plated (in } \mu\text{g)}}$$

\*cfu: colony forming unit

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 100  $\mu\text{l}$  *E. coli* cells in electroporation buffer
- 1  $\mu\text{l}$  recombinant DNA (concentration of 0.1  $\mu\text{g} / \mu\text{l}$ )
- 1  $\mu\text{l}$  sterile water
- Ice bucket with ice
- Electroporator
- Sterile cuvettes for electroporation (max. volume of 150 $\mu\text{l}$ )
- Agar plates with ampicillin
- Sterile spreader for spreading bacterial culture on agar plate
- SOC broth
- Incubator
- A variety of micropipettes, sterile micropipette tips and sterile 2 ml microcentrifuge tubes

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s showing selection of the recombinant bacteria on plates
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms

[Total: 12]

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**5 Free-response question**

Write your answers on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**(a)** Describe the polymerase chain reaction (PCR) and explain the limitations of the procedure. [8]

**(b)** Discuss the benefits and ethical concerns of the human genome project for humans. [6]

**(c)** Discuss the social implications of genetically modified crop plants. [6]

**[Total: 20]**